

TIAN SHAN, Kyrgyzstan

WMO: 369820

Lat: 41.897N

Lon: 78.185E

Elev: 3639

StdP: 64.60

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -30.5	(c) -28.4	(d) -35.8	(e) 0.2	(f) -28.2	(g) -33.6	(h) 0.3	(i) -27.0	(j) 9.3	(k) -14.9	(l) 8.2	(m) -13.9	(n) 0.6	(o) 50	(p) 0.916

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.2	(c) 14.4	(d) 6.2	(e) 12.9	(f) 5.4	(g) 11.4	(h) 4.6	(i) 8.7	(j) 11.1	(k) 7.1	(l) 10.0	(m) 5.9	(n) 9.4	(o) 4.1	(p) 0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.6	10.3	9.7	5.9	9.1	8.1	4.4	8.2	6.7	36.5	11.0	31.9	10.1	28.7	9.4	15.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.9	7.7	6.6	-33.1	18.2	2.0	2.1	-34.5	19.7	-35.7	21.0	-36.8	22.2	-38.2	23.7
			-33.5	9.4	1.8	2.7	-34.8	11.4	-35.9	12.9	-36.9	14.4	-38.3	16.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.6	-18.8	-17.0	-10.7	-4.7	-0.3	3.6	5.9	5.3	1.6	-4.7	-11.8	-16.8
	DBStd	9.40	3.97	4.32	4.55	4.21	3.04	2.19	2.28	2.24	2.94	3.69	4.43	3.66
	HDD10.0	5704	892	756	641	441	318	191	129	146	251	456	653	830
	HDD18.3	8744	1150	989	899	691	577	441	386	403	501	714	903	1088
	CDD10.0	2	0	0	0	0	0	0	2	1	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	2.2	1.5	1.7	2.0	2.4	2.5	2.6	2.6	2.4	2.6	2.4	1.8	1.7
Precipitation	PrecAvg	291	3	6	12	23	39	54	50	42	28	12	7	7
	PrecMax	439	10	21	31	59	129	104	103	96	60	44	22	22
	PrecMin	124	0	0	4	4	5	12	10	0	0	0	0	0
	PrecStd	74	3	6	6	15	24	23	23	27	16	11	6	5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.5	-3.3	2.7	8.5	11.1	13.8	16.9	16.1	12.9	8.1	1.6	-5.1
		MCWB	-9.5	-6.2	-2.1	1.6	3.4	5.1	8.4	6.5	5.4	0.1	-3.4	-7.8
	2%	DB	-9.1	-6.0	0.5	5.9	8.8	11.5	14.9	14.2	11.0	5.1	-0.9	-7.1
		MCWB	-11.0	-8.1	-3.5	0.1	2.3	4.0	6.3	6.1	3.2	-0.4	-4.3	-9.5
	5%	DB	-10.7	-8.0	-1.4	4.0	7.0	10.1	13.3	12.6	9.5	3.2	-3.0	-9.0
		MCWB	-12.1	-9.7	-4.5	-0.7	1.3	3.7	5.7	5.4	2.5	-1.5	-5.8	-10.8
	10%	DB	-12.2	-9.8	-3.0	2.1	5.2	8.7	11.7	11.1	7.9	1.4	-5.0	-10.5
		MCWB	-13.5	-11.3	-5.5	-1.7	0.6	3.2	5.2	4.7	1.9	-2.5	-7.1	-12.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.0	-5.9	-1.4	3.7	6.5	8.0	12.3	11.5	8.5	2.5	-2.1	-7.0
		MCDB	-7.1	-4.2	1.3	5.6	8.4	9.8	14.8	13.5	10.8	3.7	-0.3	-5.7
	2%	WB	-10.8	-7.9	-2.9	1.3	3.3	6.1	9.0	8.5	6.0	0.2	-3.9	-9.3
		MCDB	-9.4	-6.2	-0.3	4.5	6.6	8.4	11.3	11.2	7.8	4.0	-1.7	-7.6
	5%	WB	-12.1	-9.6	-4.2	-0.3	2.1	4.9	7.1	6.8	3.7	-1.0	-5.6	-10.7
		MCDB	-10.8	-8.1	-1.7	2.9	5.4	8.2	10.5	10.1	7.0	2.4	-3.3	-9.1
	10%	WB	-13.5	-11.2	-5.5	-1.4	1.1	3.9	5.9	5.6	2.4	-2.0	-7.0	-12.0
		MCDB	-12.2	-9.9	-3.2	1.7	4.5	7.6	10.1	9.2	6.5	0.9	-4.9	-10.5
Mean Daily Temperature Range	5% DB	MDBR	11.5	12.0	11.8	10.4	9.1	9.1	10.2	10.5	10.4	10.2	11.0	10.9
		MCDBR	12.4	12.6	12.0	12.3	11.8	11.6	13.0	13.4	13.0	12.0	12.2	12.0
		MCWBR	10.8	10.7	9.0	7.9	6.4	5.9	6.2	6.8	6.8	7.4	9.2	10.1
	5% WB	MCDBR	12.1	12.3	11.8	10.9	10.6	10.2	11.2	11.5	12.3	10.8	11.6	11.6
		MCWBR	10.7	10.7	9.5	7.9	7.1	6.9	7.5	7.7	8.7	7.7	9.2	10.0
Clear-Sky Solar Irradiance	taub		0.216	0.245	0.304	0.348	0.342	0.332	0.313	0.315	0.296	0.274	0.241	0.209
	taud		2.377	2.293	2.133	2.061	2.173	2.286	2.384	2.374	2.379	2.356	2.390	2.424
	Ebn at Noon		967	975	939	917	929	937	953	943	942	925	921	951
	Edh at Noon		78	101	136	158	146	131	117	114	105	94	77	68
All-Sky Solar Radiation	RadAvg		2.11	2.91	3.99	4.93	5.80	6.09	6.17	5.61	4.57	3.25	2.23	1.81
	RadStd		0.18	0.25	0.29	0.44	0.38	0.32	0.32	0.30	0.32	0.25	0.21	0.15

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		+0.43	N/A	N/A	N/A	-0.69	-0.85	-203	-252	+139	+72

Nomenclature: See separate page